

* $\mathbb{R}$ -scalar fields $\phi^i(x)$, $i=1, \dots, n$

$$\eta_{\mu\nu} = \eta^{\mu\nu} = \text{diag}(-1, \dots, 1)$$

in D -dim

$$\leadsto \text{On-shell: } \square \phi^i(x) = m^2 \phi^i(x)$$

which has plane wave solutions

$$\leadsto \{ (-\partial_t^2 + \partial_i \partial_i) \phi^i = m^2 \phi^i \}, \quad p_i p_i = \vec{p}^2$$

$$\Rightarrow -\partial_t^2 \tilde{\phi}^i + i^2 p_i p_i \tilde{\phi}^i = m^2 \tilde{\phi}^i$$

$$-\partial_t^2 \tilde{\phi}^i - p_i p_i \tilde{\phi}^i = m^2 \tilde{\phi}^i$$

$$\partial_t^2 \tilde{\phi}^i = -E^2 \tilde{\phi}^i$$

Since $E^2 > 0$, there is only one plane wave sol

$$\tilde{\phi}^i = \underbrace{\phi_A e^{iEt}}_{\tilde{\phi}_-} + \underbrace{\phi_B e^{-iEt}}_{\tilde{\phi}_+}$$

$$\phi^i(x) = \int \frac{d^{D-1} \vec{p}}{(2\pi)^{D-1} 2E} \left(e^{i\vec{p} \cdot \vec{x}} \right) \left(\phi_A e^{iEt} + \phi_B e^{-iEt} \right)$$

$$\phi^{i*} = \phi^i \quad (\text{Real}), \text{ thus:}$$

$$\int e^{-i\vec{p} \cdot \vec{x}} \left(\phi_A^* e^{-iEt} + \phi_B^* e^{iEt} \right) \frac{d^{D-1} \vec{p}}{(2\pi)^{D-1} 2E}$$

$$\vec{p} \rightarrow -\vec{p} \quad \int_{-\infty}^{\infty} (-1)^{D-1} d^{D-1} \vec{p} / (1) \quad ()$$

$$\text{Flip integrals} \quad (-1)^{2(D-1)} = 1$$

$\therefore$

$$\phi_A^* = \phi_B$$

$$\phi^i = \int \frac{d^{D-1} \vec{p}}{(2\pi)^{D-1} 2E} e^{i\vec{p} \cdot \vec{x}} \left(a(\vec{p}) e^{-iEt} + a^*(\vec{p}) e^{iEt} \right)$$

• Do the same for $\vec{p}$ on second term

$$\phi^i(x) = \phi_+^i(x) + \phi_-^i(x)$$

Symmetries of the system:

$$\phi^i(x) \rightarrow \phi'^i(x)$$

- If $\phi^i(x)$ is a sol. to the E.O.M then so too is $\phi'^i(x)$
- Mapping satisfies $S[\phi^i] = S[\phi'^i]$ along with appropriate B.C's.